

Module 2

⇒ Explores the challenges and opportunities presented by integrating Internet of Things (IoT) devices with fog and cloud computing

The Problem

⇒ Difficulties of using traditional cloud computing for Internet of Things (IoT) applications.

⇒ IoT involves many devices sending a lot of data from different places. Cloud computing, which relies on central data centers, can't handle this efficiently.

⇒ **Key Issues**

- **Scale and Distribution:** Too much data from too many devices can cause bottlenecks and delays.
- **Real-time Processing:** Many IoT applications need fast, real-time processing, which cloud computing can't always provide.
- **Heterogeneity and Interoperability:** Different devices have different protocols and formats, making integration challenging.
- **Limited Resources:** IoT devices have limited power, memory, and battery life, which can be strained by relying on cloud computing.

⇒ Cloud computing has played a crucial role in enabling IoT. However, the sheer number of connected devices and the data they generate will put a massive strain on current infrastructure. This will lead to difficulties in maintaining quality of service (QoS).

⇒ IoT is characterized by:

- **Extreme geographic distribution:** Devices are spread across vast areas.
- **Heterogeneity:** Devices have different protocols, formats, and capabilities.
- **Dynamism:** Devices and data are constantly changing.

⇒ These characteristics make it challenging for traditional cloud computing to efficiently handle IoT data and applications.

The Solution

- ⇒ Fog computing emerges as a middle ground between the cloud and IoT devices, **offering localized processing** and **data management** to address the limitations of cloud-only systems.
- ⇒ To address IoT requirements, a new approach is necessary. This approach must consider:
 - **Transversal requirements**: Scalability, interoperability, flexibility, reliability, efficiency, availability, and security.
 - **Cloud-to-thing (C2T) needs**: Specific computation, storage, and communication requirements for IoT devices.
- ⇒ Fog computing has the following characteristics:
 - **Intermediary compute elements**: Fog nodes are used to provide compute resources.
 - **Data management and communications services**: Fog nodes facilitate data management and communications between devices and the cloud.
 - **Enabling IoT applications**: Fog computing enables the execution of relevant IoT applications.

Analytical Models

- ⇒ These use mathematical equations to describe how the system works and predict its behavior.
- ⇒ In C2F2T, Analytical models are used to estimate things like energy consumption of IoT devices, latency in data processing, or the cost of using cloud resources
- ⇒ Analytical models are mathematical models that use equations to describe how a system works. These models can:
 - Predict how much **computing power** is **needed**
 - Measure how **changes in hardware or software** affect the system
 - **Optimize** system **performance**

Analytical models are used to study different aspects of Cloud-to-Fog-to-Thing (C2F2T) systems

- ⇒ **Modeling Architecture**: One paper described a system's architecture using vectors and equations to calculate **power consumption and latency**.
- ⇒ **Gateway Optimization**: Another paper used equations to **optimize gateway efficiency**, reducing delay and improving response time.
- ⇒ **Scheduling Mechanism**: A paper proposed an analytical model for **scheduling tasks in an IoT environment**, considering factors like processor load and memory.
- ⇒ **QoS Prediction**: A method was proposed to predict the Quality of Service (QoS) of IoT services, using a collaborative filter and equations.
- ⇒ **Scheduling Sensor Applications**: A scheduling model was proposed to manage sensor applications, considering resource requirements and priorities.

Analytical Model 1 – Scheduling Model

A scheduling model is proposed to manage sensor applications by a gateway. The model considers the resource requirements of each application.

Key Components

- ⇒ **Applications**: There are n applications, denoted as $A = \{a_i\}$.
- ⇒ **Resource Requirement Vector**: Each application a_i has a resource requirement vector R_i .
- ⇒ **Priority**: Each application a_i has a priority r_i , ranging from 0 to 1.
- ⇒ **Required Work Status**: Each application a_i has a required work status $w_i(t)$.
- ⇒ **Actual Working Status**: Each application a_i has an actual working status $s_i(t)$.

The scheduling problem is formulated using Equation, which calculates the total waiting time of tasks.

$$\begin{aligned}
 \min. \varphi &= \sum_{i=1}^n \int_0^T r_i w_i(t) [1 - s_i(t)] dt \\
 s. t. (1) \quad &\sum_{i=1}^n u_i s_i(t) \leq \alpha_c u_c \\
 (2) \quad &\sum_{i=1}^n c_i s_i(t) \leq \alpha_m c_m \\
 (3) \quad &\sum_{i=1}^n b_i^I s_i(t) \leq \alpha_I b_I \\
 (4) \quad &\sum_{i=1}^n b_i^O s_i(t) \leq \alpha_O b_O
 \end{aligned}$$

Parameters

- ⇒ **T**: Evaluation time.
- ⇒ **u_c**: Total CPU utilization.
- ⇒ **c_m**: Total memory capacity.
- ⇒ **b_I**: Input bandwidth.
- ⇒ **b_O**: Output bandwidth.
- ⇒ [**α_c**, **α_m**, **α_I**, **α_O**]: Overloading factors for CPU, memory, input, and output bandwidth.

Goal

The model aims to minimize the total waiting time of tasks while considering the resource requirements and priorities of each application.

Implication

The model indicates that the system can run smoothly to some extent, even with overloading factors. This implies that the model can help optimize resource allocation and scheduling to ensure efficient system operation.

Analytical Model 2 – Energy Consumption Model

Equation calculates the **expected energy consumption** of each device over a **monitoring period, T**:

$$E_{\text{exp}} = E[\Psi_e]g_e + i_e \int_0^{c_e E[\Psi_e]} (c_e E[\Psi_e] - \omega_e) P_e(\omega_e) d\omega_e$$

Parameters:

- ⇒ **E[Ψ_e]**: Expected query data volume of devices
- ⇒ **g_e**: Consumption rate (joule-per-bit)
- ⇒ **i_e**: "Idle" energy consumption by each device
- ⇒ **c_e**: Threshold for activating "idle" mode
- ⇒ **P_e(ω_e)**: Probability density function of query data volume

Analytical Model 3 – Cloud Billing Cost Model

Equation **calculates the expected cloud billing cost** for a group of devices:

$$B_{\text{exp}} = E[\Psi_b]g_b + i_b \int_0^{c_b} (c_b - \omega_b) P(\omega_b) d\omega_b \\ + p_b \int_{c_b}^{\infty} (\omega_b - c_b) P(\omega_b) d\omega_b$$

Parameters:

- ⇒ **E[Ψ_b]**: Expected aggregated query volume from n devices
- ⇒ **g_b**: Transfer/storage cost (dollars-per-bit)
- ⇒ **i_b**: Cost of transferring one bit (dollars-per-query-bit)
- ⇒ **c_b**: Coupling point defining expected billing cost

- ⇒ The energy consumption model considers both **active and idle modes**.
- ⇒ The cloud billing cost model accounts for **transfer/storage costs, idle billing costs, and active billing costs**.
- ⇒ Both models rely on **probability density functions** to capture uncertainty in query data volumes

Analytical Model 4 – Energy Consumption Model

The strategy uses **fog computing with microgrids** to reduce energy consumption of IoT applications.

Two equations are proposed to evaluate energy consumption:

$$E_{IoT-cloud} = E_{GW-r} + E_{GW-t} + E_{net} + E_{DC}$$

Parameters:

- ⇒ **EGW-r**: Energy consumed by IoT gateways when receiving data from IoT devices and sensors
- ⇒ **EGW-t**: Energy consumed by IoT gateways to transmit data to the cloud data center
- ⇒ **Enet**: Energy consumed by transport network between IoT gateways and cloud
- ⇒ **EDC**: Energy consumed by components of data center

- ⇒ The energy consumption model considers the energy consumed by **IoT gateways, transport networks, and data centers**.
- ⇒ The strategy aims to reduce energy consumption by using fog computing with microgrids.
- ⇒ Equation provides a comprehensive view of energy consumption in cloud computing-based IoT applications.

$$E_{IoT-fog} = E_{GW-r} + E_{GW-c} + \beta(E_{GW-t} + E_{net} + E_{DC})$$

Parameters:

- ⇒ **EGW-r**: Energy consumed by IoT gateways when receiving data from IoT devices and sensors
- ⇒ **EGW-t**: Energy consumed by IoT gateways to transmit data to the fog
- ⇒ **EGW-c**: Energy consumed by IoT gateways for local computation and processing
- ⇒ **Enet**: Energy consumed by transport network between IoT gateways and fog/cloud

- ⇒ **EDC**: Energy consumed by components of data center
- ⇒ **β** : Ratio of the number of updates from the fog to the cloud for synchronization
- ⇒ Equation calculates the **energy consumption of communication between IoT and fog**.
- ⇒ The equation **considers local computation and processing energy** (EGW-c) and **synchronization energy** (β).
- ⇒ The energy consumption is reduced by processing data locally in the fog and minimizing updates to the cloud.

Analytical Model 5 – Reliability Model

The proposed model **estimates the reliability of an IoT scenario** in a smart home context. The IoT system consists of n subsystems.

$$R_s(t_b) = \prod_{i=1}^N ISR_i \times \prod_{i=1}^f P_f(i) \times \prod_{i=1}^k P_{pr}(i)$$

Parameters:

- ⇒ **Ppr**: Availability of programs running on virtual machines
- ⇒ **k**: Number of programs
- ⇒ **Pf**: Availability of input files for programs
- ⇒ **f**: Number of input files
- ⇒ **ISRi**: Reliability of each subsystem (virtual machine)
- ⇒ **n**: Number of subsystems
- ⇒ The model considers the **availability of programs, input files, and subsystem reliability**.
- ⇒ The model provides a comprehensive approach to estimating IoT reliability in smart home scenarios.

Other Models

Security Evaluation

Study [15]: Presented a model to evaluate the security level of C2T systems, focusing on information flow and system states.

Optimization Problems

Study [23]: Used a knapsack problem (MKP) to optimize service allocation in C2F2T scenarios, considering load balancing, delay, and energy consumption.

Study [32]: Formulated an optimization problem to maximize cloud service utilities, obtaining optimal payment and computation offloading.

Study [11]: Proposed analytical modeling to represent QoS of IoT composite services, optimizing cost with QoS constraints.

Architecture Comparison

Study [31]: Used analytical models to decide whether offloading computing should be processed in IoT devices or the cloud, considering QoS and energy levels.

Study [27]: Compared the performance of fog architecture against traditional cloud computing, considering device location, operation mode, hardware details, and event types.

Mobile Nodes and Cloud Computing

Study [12]: Modeled mobile devices connected to cloud computing, considering device movement and cellular connections.

Study [19]: Presented a model detailing the architecture of mobile nodes connected to cloud computing, including WSN, cloud infrastructure, applications, and mobile users.

Integrated C2F2T literature by Use-case

Resource Management:

- **Energy Consumption:** Many studies focus on models that represent applications in health use case scenarios, such as medical research, health monitoring, and wearable IoT architectures for healthcare systems.
- **Resource Allocation:** Some papers address resource allocation in a C2F2T context, considering aspects like load balancing, delay, and energy consumption.
- **Quality of Service (QoS):** QoS in IoT and cloud-IoT environments is discussed, including the use of mathematical models to calculate QoS of IoT services and neighborhood models for QoS predictions.

Smart Cities:

- **Crowdsensing Applications:** These applications are explored in smart traffic scenarios and air quality sensor applications using mobile sensors and devices.
- **Smart Living:** Use cases include smart living spaces with gateways to access cloud resources, fire alarm systems, and airplane monitoring scenarios.

Wireless Sensor Networks (WSN):

- **Sensing as a Service:** WSN models for sensing as a service that integrate IoT and cloud infrastructures.
- **Fault Tolerance:** A fault fuzzy-ontology is proposed to verify fault tolerance in large-scale WSN using service-oriented applications.

Generic IoT/Fog/Cloud Applications:

- Three-tier models addressing the C2F2T scenario.
- Models to improve response time of IoT systems using fog computing as an intermediary platform.
- Generic fog architectures.
- Architectures focused on offloading computing to the edge.

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Energy Consumption	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
Resource Management	Energy Efficiency of IoT Devices	Analytical Model	Energy Consumption	Improve energy efficiency of IoT devices.
Resource Management	Energy Consumption of WSN	Analytical Model	Energy Consumption	Analyze energy consumption in a WSN.
Resource Management	Energy Consumption in Fog-IoT Integration	Analytical Model	Energy Consumption	Reduce energy consumption by integrating IoT and fog computing.
Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Energy Consumption, Resource Usage	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Response Time, Reliability, Availability	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
Smart Cities	Crowdsensing in Smart Traffic	Petri Net	System Effectiveness	Evaluate a mobile crowdsensing-as-a-service architecture.

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Smart Cities	Air Quality Sensing with Mobile Devices	Analytical Model	Energy Consumption, Percentage Energy Saved	Model energy consumption in a mobile crowdsensing application.
Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	QoS (Object Detection Accuracy), Energy Consumption	Determine optimal processing location (edge or cloud) for video data.
Smart Cities	Smart Living Spaces with Cloud Access	Analytical Model	Scheduling Performance (Scalability, Waiting Time, Throughput)	Develop a scheduling model for managing sensor applications in a smart living space.
Smart Cities	Fire Alarm System	Analytical Model	Reliability	Evaluate the reliability of a fire alarm system in an IoT context.
Smart Cities	Airplane Monitoring	Bayesian Model	-	Predict flight delays based on equipment data.
Wireless Sensor Networks (WSN)	Sensing as a Service	Fault Fuzzy-Ontology	-	Verify fault tolerance in large-scale WSN.
Wireless Sensor Networks (WSN)	Fault Tolerance	Analytical Model	Packet Delivery Latency	Model a WSN for sensing as a service integrated with IoT and cloud.
Health	Health Monitoring	Analytical Model	-	Develop a framework for sharing IoT devices in health monitoring.
Health	Medical Research	Analytical Model	Security (Opacity)	Analyse security of information flow in cloud-integrated IoT systems.

Use Case Scenario	Application	Modelling Technique	Metrics	Key Findings/Objectives
Health	Wearable IoT for Healthcare	Petri Net	Reliability	Manage traceability of data in a wearable IoT architecture.
Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Analytical Model	Energy Consumption, Latency	Represent a generic C2F2T architecture and analyze energy consumption and latency.
Generic IoT/Fog/Cloud	Fog Computing for Improved Response Time	Analytical Model	Effective Arrival Rate, Average System Response, Average Buffer Occupancy	Improve response time of IoT systems using fog computing.
Generic IoT/Fog/Cloud	Generic Fog Architecture	Analytical Model	Energy Consumption, Latency	Compare performance of fog architecture against traditional cloud computing.
Generic IoT/Fog/Cloud	Edge Offloading	Analytical Model	Resource Consumption (Cloud and IoT Device Consumption)	Improve mobile user experience by offloading computation to the edge.

Integrated C2F2T Literature by Metric

Energy Consumption:

This is the most frequently evaluated metric, appearing in 10 of the analyzed articles.

Studies focus on:

- Energy consumption in different layers of the C2F2T architecture (IoT, fog, and cloud).
- Energy consumption of devices in active, idle, and transition states.
- Energy consumption related to data transmission, reception, sensing, and computing.
- Impact of communication distance and medium on energy consumption.
- Energy consumption in relation to data flow and synchronization between different layers.
- Energy consumption of fog devices powered by renewable energy sources.
- Minimizing energy consumption in C2F2T architectures.

Performance:

This metric focuses on evaluating the efficiency and responsiveness of C2F2T systems.

Studies assess:

- Latency in processing and data transmission.
- Packet delivery latency in WSNs.
- Delay in processing application requests.
- Performance of scheduling mechanisms.
- Effectiveness of mobile crowdsensing systems.
- Response time of gateways and data processing efficiency.

Resource Consumption:

This metric examines the efficient utilization of resources in C2F2T environments.

Studies investigate:

- Optimal allocation of services and resources across devices.
- Resource consumption in cloud infrastructure and IoT devices.
- Buffer occupancy of gateways to handle data flow.

Cost:

This metric analyzes the financial implications of deploying and operating C2F2T systems. Studies evaluate:

- Billing costs associated with cloud resource usage.
- Operational costs of provisioning IoT traffic to the cloud.

Quality of Service (QoS):

This metric focuses on ensuring reliable and satisfactory performance of C2F2T applications. Studies assess:

- QoS of composite services in IoT environments.
- QoS in terms of object detection accuracy in video streams.
- Reliability of IoT systems and traceability models.

Security:

This metric addresses the security aspects of C2F2T systems, with a focus on: Opacity in data flow to protect sensitive information.

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Energy Consumption	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
Energy Consumption	Resource Management	Energy Efficiency of IoT Devices	Analytical Model	Improve energy efficiency of IoT devices.
Energy Consumption	Resource Management	Energy Consumption of WSN	Analytical Model	Analyze energy consumption in a WSN.
Energy Consumption	Resource Management	Energy Consumption in Fog-IoT Integration	Analytical Model	Reduce energy consumption by integrating IoT and fog computing.
Energy Consumption	Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Energy Consumption	Smart Cities	Air Quality Sensing with Mobile Devices	Analytical Model	Model energy consumption in a mobile crowdsensing application.
Energy Consumption	Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	Determine optimal processing location (edge or cloud) for video data.

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Energy Consumption	Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Analytical Model	Represent a generic C2F2T architecture and analyze energy consumption.
Energy Consumption	Generic IoT/Fog/Cloud	Generic Fog Architecture	Analytical Model	Compare performance of fog architecture against traditional cloud computing.
Performance	Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
Performance	Smart Cities	Crowdsensing in Smart Traffic	Petri Net	Evaluate a mobile crowdsensing-as-a-service architecture.
Performance	Smart Cities	Smart Living Spaces with Cloud Access	Analytical Model	Develop a scheduling model for managing sensor applications in a smart living space.
Performance	Wireless Sensor Networks (WSN)	Fault Tolerance	Analytical Model	Model a WSN for sensing as a service integrated with IoT and cloud.
Performance	Generic IoT/Fog/Cloud	Fog Computing for Improved Response Time	Analytical Model	Improve response time of IoT systems using fog computing.

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Resource Consumption	Resource Management	Resource Allocation in C2F2T	Analytical Model, Integer Linear Programming	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.
Resource Consumption	Generic IoT/Fog/Cloud	Edge Offloading	Analytical Model	Improve mobile user experience by offloading computation to the edge.
Cost	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Analytical Model	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
QoS	Resource Management	QoS in IoT and Cloud-IoT	Analytical Model	Calculate QoS of IoT services and predict QoS for IoT and cloud services.
QoS	Smart Cities	Vehicle-to-Cloud Monitoring	Analytical Model	Determine optimal processing location (edge or cloud) for video data.
QoS	Smart Cities	Fire Alarm System	Analytical Model	Evaluate the reliability of a fire alarm system in an IoT context.
QoS	Health	Wearable IoT for Healthcare	Petri Net	Manage traceability of data in a wearable IoT architecture.

Metric	Use Case Scenario	Application	Modeling Technique	Key Findings/Objectives
Security	Health	Medical Research	Analytical Model	Analyze security of information flow in cloud-integrated IoT systems.

Integrated C2F2T Literature by Modeling Technique

Analytical Models:

These are mathematical models with closed-form solutions, used to predict computing resource requirements and measure the effects of hardware and software changes. They dominate the C2F2T literature, with 16 articles using them to:

1. Define architectures and calculate metrics like power consumption and latency.
2. Model the number of gateways, data volume, and processing time.
3. Represent scheduling mechanisms in IoT environments.
4. Predict QoS of IoT services based on latency, response time, and network conditions.
5. Manage sensor applications by a gateway, considering resource requirements.
6. Represent energy consumption of devices and cloud billing methods.
7. Evaluate energy consumption in IoT-fog-cloud communication.
8. Estimate the reliability of IoT scenarios in smart homes.
9. Evaluate the security level of C2T systems with respect to information flow.
10. Solve optimization problems related to service allocation, cost, and energy consumption.
11. Compare performance of different architectures and offloading strategies.
12. Represent mobile nodes connected to cloud computing.

Petri Nets:

These are graphical models used to represent and analyze concurrent systems. Two articles in the document utilize Petri Nets to:

1. Model and compare a proposed mobile crowdsensing framework with standard architectures.
2. Evaluate the management of data traceability in a C2T scenario with a wearable IoT architecture.

Integer Linear Programming (ILP):

This is a mathematical optimization technique used to solve problems with integer variables. Two articles employ ILP to:

1. Calculate the financial cost of a fog-cloud architecture in a metropolitan area network.
2. Model the energy consumption of an IoT architecture composed of mini clouds.

Other Approaches:

These include Markov chains, Bayesian models, and fuzzy logic, used in specific cases to:

1. Model energy consumption of an IoT-based system for collision detection.
2. Predict events in an IoT application connected to the cloud.
3. Analyze faults in WSN scenarios by combining ontology with fuzzy logic.

Modelling Technique	Use Case Scenario	Application	Metrics	Key Findings/Objectives
Analytical Model	Resource Management	Energy Consumption of Devices in Cloud-IoT Context	Energy Consumption	Analyze the relationship between energy consumption of IoT devices and cloud billing costs.
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Analytical Model	Smart Cities	Fire Alarm System	Reliability	Evaluate the reliability of a fire alarm system in an IoT context.
Analytical Model	Wireless Sensor Networks (WSN)	Fault Tolerance	Packet Delivery Latency	Model a WSN for sensing as a service integrated with IoT and cloud.
Analytical Model	Health	Health Monitoring	-	Develop a framework for sharing IoT devices in health monitoring.
Analytical Model	Health	Medical Research	Security (Opacity)	Analyze security of information flow in cloud-integrated IoT systems.
Analytical Model	Generic IoT/Fog/Cloud	Three-tier C2F2T Model	Energy Consumption, Latency	Represent a generic C2F2T architecture and analyze energy consumption and latency.
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Analytical Model	Generic IoT/Fog/Cloud	Generic Fog Architecture	Energy Consumption, Latency	Compare performance of fog architecture against traditional cloud computing.
Analytical Model	Generic IoT/Fog/Cloud	Edge Offloading	Resource Consumption (Cloud and IoT Device Consumption)	Improve mobile user experience by offloading computation to the edge.
Petri Net	Smart Cities	Crowdsensing in Smart Traffic	System Effectiveness	Evaluate a mobile crowdsensing-as-a-service architecture.
Petri Net	Health	Wearable IoT for Healthcare	Reliability	Manage traceability of data in a wearable IoT architecture.
Integer Linear Programming (ILP)	Resource Management	Resource Allocation in C2F2T	Energy Consumption, Resource Usage	Optimize resource allocation considering energy consumption, processing capacity, and overall service allocation.

Other Approaches	Smart Cities	Airplane Monitoring	-	Predict flight delays based on equipment data.
Other Approaches	Wireless Sensor Networks (WSN)	Fault Tolerance	-	Verify fault tolerance in large-scale WSN using service-oriented applications.